

SUBJECT	Lower Secondary Maths Year 8
TOPIC	Module One
ASSIGNMENT	One
STUDENT NAME	

FEEDBACK AND REFLECTION:

As this is your first assignment, you do not have feedback from previous work to consult before you begin. But in future assignments we will encourage you to remind yourself of the feedback on your previous piece of work. Why do you think this will be important? Tick all that apply to you:

- ☐ I will be able to build on strengths in my work that my tutor has identified.
- ☐ I will avoid making the same mistakes that I made before.
- ☐ I will have a clearer idea of what I need to aim for in my work.
- ☐ I will feel more confident about my work.
- ☐ I will be able to act on my tutor's advice and so make progress.

What matters most to you in the feedback you receive from your tutor? Let us know here:

ASSIGNMENT GUIDANCE

- Make sure you write your full name clearly in the box at the top of this page.
- Please **write** your answers **by hand** in **black ink** on this question paper unless you have obtained specific permission to submit typed work.
- This is not a timed assignment. We estimate that this assignment may take you between 30 and 40 minutes. However, if you need to take longer, then please do so.
- You are not expected to complete your assignments without referring to your notes or coursebook.
- In order for maximum marks to be awarded, for any question worth more than **one mark**, please show your workings to make your method clear.
- Calculators are **not** to be used in this assignment.





LS Maths Assignment One

1. Put these numbers in order from smallest to largest:

a. $-2, -7, 0, 9, -17$ (2 marks)

b. $-1.7, -4.2, 0.9, -0.3, 12.7$ (2 marks)

(Total 4 marks)

2. Calculate:

a. $-5 - -2$ (1 mark)

b. $-17 + -21$ (1 mark)

c. $3 - +7$ (1 mark)

d. $-2 - 10$ (1 mark)

(Total 4 marks)

3. Calculate:

a. 8×-4 (1 mark)

b. $-11 \div 2$ (1 mark)

c. $-72 \div -9$ (1 mark)





d. -4×-6

(1 mark)

(Total 4 marks)

4. The temperature outside is -4°C and overnight it falls further by 6°C .

a. What is the temperature overnight?

(1 mark)

b. Inside the building, the temperature remains steady at 21°C . What is the temperature difference between inside and outside overnight? Remember to make your method clear.

(2 marks)

(Total 3 marks)

5. Fill in the spaces in these Fibonacci style sequences:

a. 5, 20, _____, 45, 70

(1 mark)

b. 0.2, 0.4, 0.6, _____, 1.6

(1 mark)

c. -5, -1, -6, -7, _____

(1 mark)

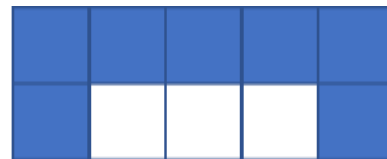
(Total 3 marks)





6.

a. Draw the next two diagrams in the sequence.



(2 marks)

b. Complete this table:

Number of white tiles					
Number of blue tiles					

(1 mark)

c. Describe the **term-to-term** rule for the number of blue tiles. Be sure to give enough information to describe the sequence fully, by stating the first term and the common difference.

(2 marks)

d. Describe the **position-to-term** rule for the number of blue tiles.

(2 marks)
(Total 7 marks)





7. A Wolsey Hall student plants a flower bush with five of the buds already flowered. Each week another three buds flower, and the student wishes to know how many in total have flowered by the end of each week.

a. Explain why this sequence starts 8,11,14 rather than 5,8,11....

(1 mark)

b. How many buds will have flowered after three weeks?

(1 mark)

c. If n represents the number of weeks since the student planted the bush, write a rule to represent how many buds will flower after n weeks

(2 marks)

d. How many weeks will it take for 32 buds to have flowered?

(1 mark)

(Total 5 marks)





8.

- a. Find the **term-to-term** rule for each of these linear sequences. Be sure to give enough information to describe the sequences fully, by stating the first term and the common difference.

i. 16, 21, 26, 31, 36 ... (2 marks)

ii. 75, 66, 57, 48, 39 ... (2 marks)

- b. Now consider the **position-to-term** rule for each sequence in part a. Can you find a general expression for the n th term?

i. (2 marks)

ii (2 marks)





- c. Show how you can use your general term, found in part c), to find the 10th term of each sequence. Remember to show your method clearly!

i. (1 mark)

ii. (1 mark)

(Total 10 marks)

TOTAL FOR ASSIGNMENT 40 MARKS

